Idea: configuration space for dynamical system

(ex: classical mechanics : positions + velocities (if Q space of position $\to TQ$)

Lagrangian is a function of (possibly) time and of configuration space. This function encapsulates dynamics $\to$ it gives equations of motion

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In practice: if $\mathcal{L} = \mathcal{L}(q,t) \in \mathbb{R}$ $\quad q \in M$ (configuration space)

to each path in configuration space $\gamma : t \in [a,b] \to \gamma(t) \in M$

we associate a number: $$S[\gamma] = \int_a^b \mathcal{L}(\gamma(t), t)\, dt \qquad \text{(action)}$$

$\to$ physical path are those that "minimize" the action

$\to$ extremize

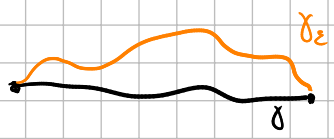
variation of S?

take γ build $\gamma_\varepsilon : [a,b] \to M$ such that $\gamma_0(t) = \gamma(t) \quad \forall t$

$\varepsilon \in (-1,1)$ $\qquad \gamma_\varepsilon(a) = \gamma(a) \quad \forall \varepsilon$

$\gamma_\varepsilon(b) = \gamma(b) \quad \forall \varepsilon$

map $\gamma \to \gamma_\varepsilon$ smooth in ε



define variation of S as $$\delta S[\gamma] = \frac{d}{d\varepsilon} S[\gamma_\varepsilon]\Big|_{\varepsilon=0}$$

γ is extremal if $\delta S[\gamma] = 0$ for all deformations of γ

This leads to Euler-Lagrange eqs.

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coordinates on M $\subset_{\text{open}} \mathbb{R}^{2n}$: (q, v) $\qquad q, v$ n-dimensional $\qquad q = (q^1, q^2, \dots q^n)$

suppose $\mathcal{L} = \mathcal{L}(q, v, t)$

$\gamma(t) = (q(t), \dot{q}(t))$

$$\rightarrow \quad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial v^\mu}(q(t),\dot{q}(t),t)\right) = \frac{\partial \mathcal{L}}{\partial q^\mu}(q(t),\dot{q}(t),t)$$

in classical mechanics $\quad \mathcal{L} = \underset{\text{kinetic}}{T} - V_{\text{potential}} \quad \rightarrow$ Newton's law

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<u>Fields</u> $\quad \varphi : M \rightarrow \mathbb{R}$ scalar field $\quad M$ spacetime (n-dimensional)

$$\mathcal{L} = \mathcal{L}(\underset{\text{number}}{\varphi}, v, \overset{\in M}{x}) \qquad \mathcal{L} : \mathbb{R} \times \mathbb{R}^n \times M \rightarrow \mathbb{R}$$

$x \in M$

v represents values of $\nabla\varphi$

Minkowski:

<u>ex:</u> massless free scalar field $\quad \mathcal{L}(\varphi, \partial\varphi, x) = \frac{1}{2}\eta^{\mu\nu}\partial_\mu\varphi\,\partial_\nu\varphi = -\frac{1}{2}(\partial_0\varphi)^2 + \frac{1}{2}((\partial_1\varphi)^2 + \dots)$

Euler-Lagrange eqs. $\quad \sum_\mu \frac{\partial}{\partial x^\mu}\left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \varphi}(\varphi(x), \partial\varphi(x), x)\right) = \frac{\partial \mathcal{L}}{\partial \varphi}(\varphi(x), \partial\varphi(x), x)$

sum over μ

rule: whenever there are two indices in the same side of an eq. one is up and one is down, there is an implicit sum

x^μ up $\quad \frac{\partial}{\partial x^\mu}$ down $\quad \frac{\partial}{\partial \partial_\mu \varphi}$ up $\quad \partial_\mu\varphi$ down $= \frac{\partial}{\partial x^\mu}\varphi \quad x^\mu p_\mu$

To be precise $\quad \mathcal{L}(\varphi, v, x) = \frac{1}{2}\eta^{\mu\nu} v_\mu v_\nu$

E.L: $\quad \frac{\partial}{\partial x^\mu}\left(\frac{\partial \mathcal{L}}{\partial v_\mu}(\varphi(x), \partial\varphi(x), x)\right) = \frac{\partial \mathcal{L}}{\partial \varphi}(\varphi(x), \partial\varphi(x), x)$

$$\frac{\partial \mathcal{L}}{\partial \varphi} = 0 \qquad \frac{\partial \mathcal{L}}{\partial v_\mu} = \frac{1}{2}\eta^{\mu\nu} v_\nu + \frac{1}{2}\eta^{\nu\mu} v_\nu = \eta^{\mu\nu} v_\nu \qquad c = 1$$

evaluate at $v = \partial\varphi$

$$\frac{\partial}{\partial x^\mu}\left(\eta^{\mu\nu}\partial_\nu\varphi\right) = 0 \rightarrow \boxed{\eta^{\mu\nu}\partial_\mu\partial_\nu\varphi = 0} \rightarrow +\frac{\partial^2\varphi}{\partial t^2} - \frac{\partial^2\varphi}{\partial x^2} - \frac{\partial^2\varphi}{\partial y^2} - \frac{\partial^2\varphi}{\partial z^2} = 0$$

Wave eq

Physicist way: $\quad \mathcal{L} = \frac{1}{2}\eta^{\mu\nu}\partial_\mu\varphi\,\partial_\nu\varphi \qquad \frac{\partial \mathcal{L}}{\partial \partial_\mu \varphi} = \eta^{\mu\nu}\partial_\nu\varphi$

$$-\frac{\partial^2\varphi}{\partial t^2} + \nabla^2\varphi = 0$$ free scalar field no mass

More generally $\mathcal{L} = \frac{1}{2}\left(\eta^{\mu\nu}\partial_\mu\varphi\,\partial_\nu\varphi - m^2\varphi^2\right)$ Klein-Gordon Lagrangian

$$\frac{\partial\mathcal{L}}{\partial\varphi} = -m^2\varphi \quad \rightarrow \quad \partial^\mu\partial_\mu\varphi + m^2\varphi = 0$$ if $\eta = \mathrm{diag}(1,-1,-1,-1)$

$$\left(\frac{\partial^2}{\partial t^2} - \nabla^2 + m^2\right)\varphi = 0$$

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add interactions: $\mathcal{L}(\varphi,\partial\varphi) = \frac{1}{2}\left(\partial^\mu\varphi\,\partial_\mu\varphi - m^2\varphi^2\right) - \overbrace{V(\varphi)}^{\text{self-interaction part}}$

Lagrangian for electromagnetism

idea: we have $E = E(t,x)$ electric, $B = B(t,x)$ magnetic field

$$\boxed{\begin{aligned}\nabla\cdot B &= 0\\ \nabla\times E &= -\frac{\partial B}{\partial t}\end{aligned}} \qquad \begin{aligned}\nabla\cdot E &= 0\\ \nabla\times B &= \frac{\partial E}{\partial t}\end{aligned}$$

$F = -\nabla V$

$$\nabla\cdot B = 0 \quad \Rightarrow \quad B = \nabla\times A$$

$$\nabla\times E = -\nabla\times\left(\frac{\partial A}{\partial t}\right) \Rightarrow \nabla\times\left(E + \frac{\partial A}{\partial t}\right) = 0 \quad \Rightarrow \quad E + \frac{\partial A}{\partial t} = -\nabla\phi$$

$$E = -\nabla\phi - \frac{\partial A}{\partial t}$$

ϕ and A scalar and vector potential

in diff. geometry: E 1-form B 2-form $dB = 0$ $dE = -d\left(\frac{\partial A}{\partial t}\right)$

$\downarrow$ $B = dA$ A 1-form

$$\nabla\cdot E = 0 \quad \rightarrow \quad -\nabla^2\phi - \frac{\partial}{\partial t}\nabla\cdot A = 0$$

$$\nabla\times B = \frac{\partial E}{\partial t} \quad \rightarrow \quad \nabla(\nabla\cdot A) - \nabla^2 A = -\nabla\frac{\partial\phi}{\partial t} - \frac{\partial^2 A}{\partial t^2}$$

$\nabla\times\nabla\times A = \nabla(\nabla\cdot A) - \nabla^2 A$

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gauge fixing A, ϕ are not fully specified

$E = -\nabla\phi - \frac{\partial A}{\partial t}$

if $A \rightarrow A + \nabla\lambda$, $\phi \rightarrow \phi - \frac{\partial\lambda}{\partial t}$ gauge transformation

$B \rightarrow B$

$$E \rightarrow -\nabla\left(\phi - \frac{\partial\lambda}{\partial t}\right) - \frac{\partial}{\partial t}\left(A + \nabla\lambda\right)$$

$$= E$$

Lorenz gauge : $\boxed{\nabla \cdot A + \frac{\partial \varphi}{\partial t} = 0}$ $\qquad \nabla \cdot A = -\frac{\partial \varphi}{\partial t}$

$$-\nabla^2 \phi - \frac{\partial}{\partial t} \nabla \cdot A = 0 \rightarrow -\nabla^2 \phi + \frac{\partial^2 \phi}{\partial t^2} = 0$$

$$\nabla(\cancel{\nabla} \cdot A) - \nabla^2 A = -\cancel{\nabla \frac{\partial \phi}{\partial t}} - \frac{\partial^2 A}{\partial t^2} \rightarrow -\nabla^2 A + \frac{\partial^2 A}{\partial t^2} = 0$$